ASHISH KOSANA

Software Engineer, Backend and Full-Stack
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SUMMARY

Backend and full-stack software engineer. I build and ship features end to end, from Amazon Web Services (AWS) infrastructure as code and REST APIs secured with Cognito to cross-platform Flutter and TypeScript user interfaces, deployed to production with typed tests that pass in Continuous Integration (CI). Strong in data structures, algorithms, and systems fundamentals, and I use LLMs to build and ship faster.

SKILLS

- **Languages:** Python, TypeScript, JavaScript, Dart, SQL, Bash, Linux
- **Backend and APIs:** REST, HTTP, JSON API design, FastAPI, SQLAlchemy, pydantic v2, OAuth 2.0, JSON Web Tokens (JWT), Stripe, hexagonal (ports and adapters) architecture, caching, microservices
- **Cloud and Infrastructure (AWS):** Lambda, API Gateway, Cognito, CDK for Infrastructure as Code (IaC), EventBridge, Secrets Manager, IAM, S3
- **Databases:** DynamoDB (NoSQL), PostgreSQL, SQLite (relational database) **Containers:** Docker, docker-compose
- **Frontend:** Next.js, React (server-side rendered Progressive Web App, PWA), Flutter (cross-platform iOS, Android)
- **Testing:** pytest, unit and integration testing, test doubles and fakes, mypy --strict, ruff, evaluation harness (precision/recall/F1)
- **Workflow:** Git, pull request (PR) and code review, Continuous Integration/Continuous Deployment (CI/CD) with GitHub Actions, Agile
- **Core CS:** data structures, algorithms, Object-Oriented Programming (OOP), operating systems, computer networks

EXPERIENCE

Software Engineer (Intern, full-time conversion track) | Crewtron **|** Tempe, AZ **|** May 2026 - Present

- Shipped clock-in/out with geofence auto-tracking and a live crew map end to end, using a Flutter UI, Python on AWS Lambda, and DynamoDB. Crew visibility is scoped by role, and presence triggers auto clock-in.
- Designed REST APIs secured with AWS Cognito and provisioned the backend as code with AWS CDK and scoped IAM policies. Built merchant review-request flows over email/SMS.
- Built cross-platform Flutter features on the Stripe API (invoicing, quotes, payments) and a UTC timezone pipeline that drives a timezone-aware calendar.
- Improved quality with structured QA, OWASP Top 10 security reviews, and a consistent Git/PR workflow. Authored an API integration-test suite reaching **~82% coverage** and integrated LLM-based automation into internal tooling.

PROJECTS

snip | URL-shortener API (pure backend/systems) | FastAPI, SQLAlchemy, SQLite, Docker, no LLM
github.com/Ashishkosana/snip

- Generates short links with a base62 codec that has no dependencies and avoids collisions. A StorePort protocol lets the SQLite and in-memory backends swap in behind one interface, so the API layer never touches a concrete store.
- Adds a per-IP token-bucket rate limiter (429 on exceed) and TTL link expiry (410). **28 tests** cover the codec, rate limiting, and expiry, with mypy --strict, Docker, and CI green.

Rythu | Telugu-first farmer weather and crop-advisory PWA (deployed) | Next.js, TypeScript, Python, AWS Lambda, DynamoDB, API Gateway, CDK, Amplify
github.com/Ashishkosana/rythu

- Caches third-party weather forecasts in DynamoDB with a TTL, so repeat lookups skip the external weather API. This cuts latency and cost.
- Ships a server-rendered Telugu PWA (Next.js on AWS Amplify, git-push CI/CD) over a Python Lambda and API Gateway backend, with infrastructure as code in AWS CDK. A rule-based advisory engine uses GPS and OpenStreetMap geocoding and applies government-verified fertilizer rules that account for nutrient overlap and prevent over-application. **~135 tests** (pytest and Vitest), mypy --strict, CI green.

review-lens | code-review tool that runs multiple review lenses and verifies its own findings | Python, pydantic v2, CLI and GitHub Action
github.com/Ashishkosana/review-lens

- Runs an LLM across multiple independent review lenses, then a verification pass rechecks each finding before it is reported. This favors precision over recall to cut false positives.
- Includes a precision/recall/F1 evaluation harness over a labeled diff dataset, and ships as both a CLI and a GitHub Action. **74 tests**, mypy --strict, CI green.

EDUCATION

Bachelor of Science (B.S.) in Computer Science | University of Massachusetts Lowell **|** Graduated Dec 2025
GPA 3.51/4.00 **|** Dean's List **|** Transferred from SR Institute of Technology (JNTU), Hyderabad to complete the B.S.